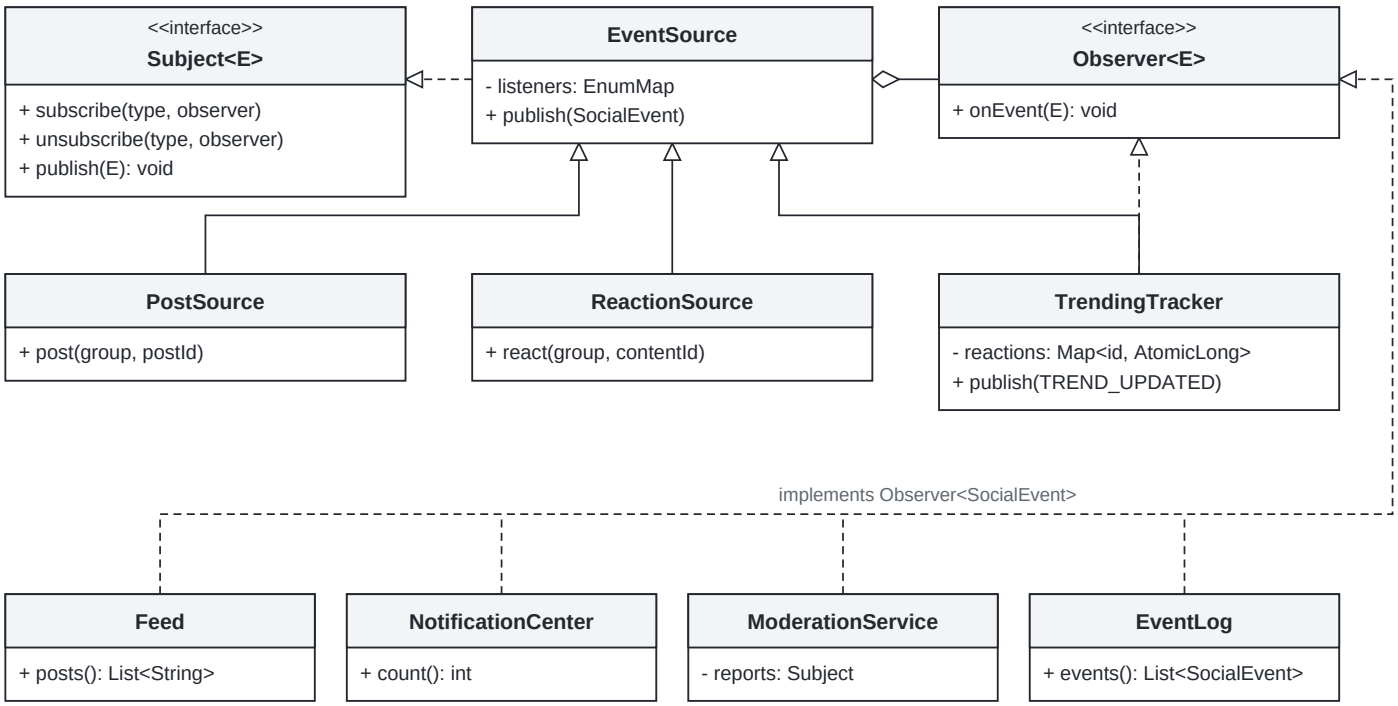
Software Design Patterns

Assignment 6

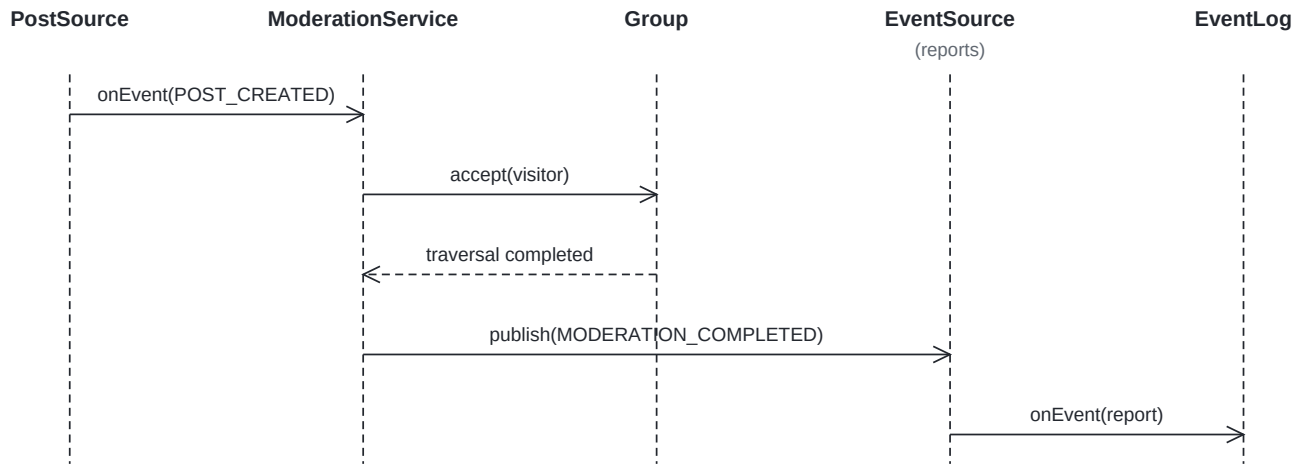
by Sabirzhanov Emil, SE-2533

Observer Class Diagram



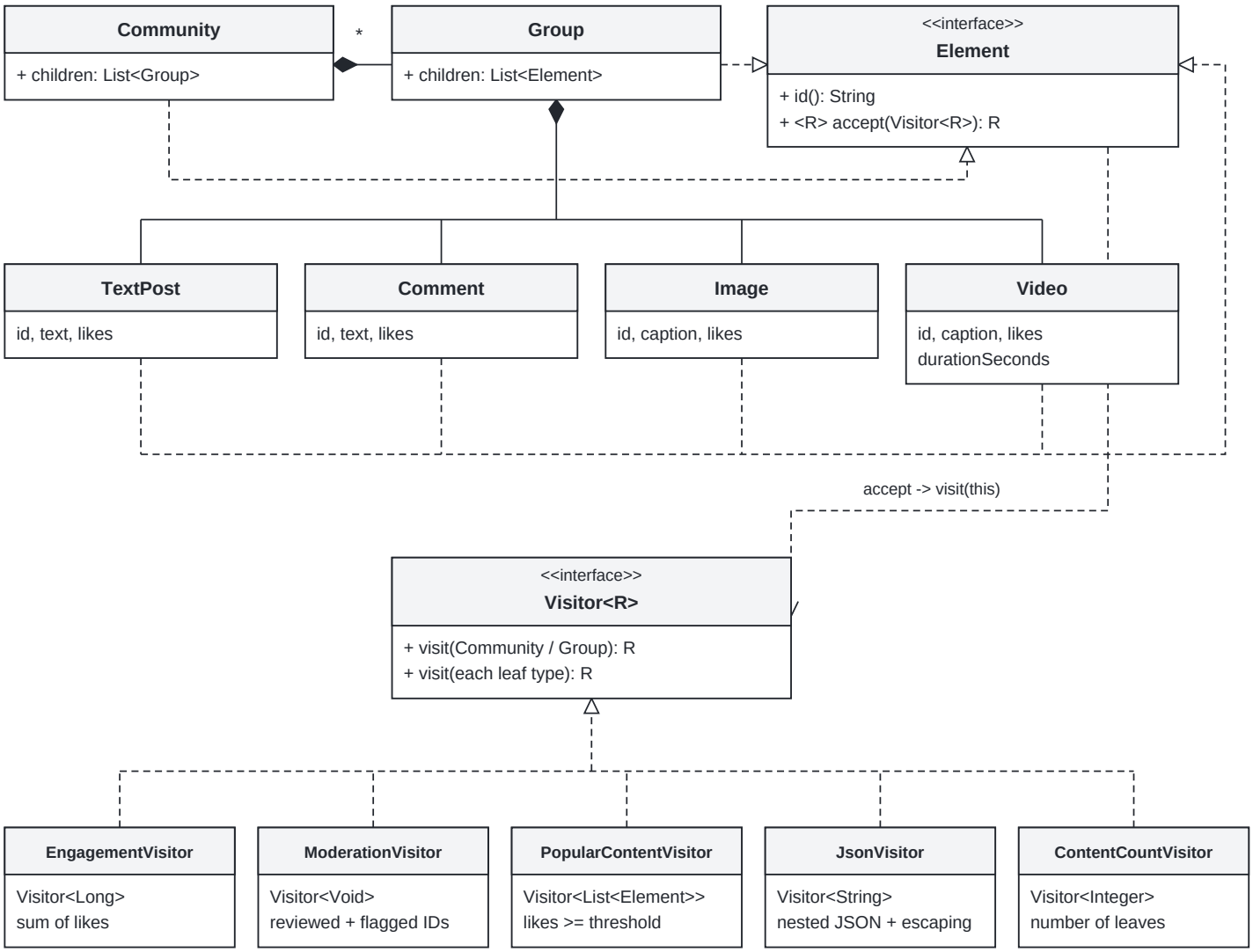
Push delivery; subscriptions by event type; snapshot listener registry.

Moderation Sequence Diagram



Only the event group is visited. Each event gets its own ModerationVisitor.

Composite and Visitor Class Diagram



Variant 16 - Social Network

Immutable tree: Community -> Group -> posts, comments, images and videos.

Visitors own traversal. Element.accept calls visitor.visit(this) for double dispatch.

Engagement: 10 + 2 + 5 + 7 + 3 = 27; 5 leaves; flagged content: c1.

30 JUnit 5 tests passed, including 4 threads x 1,000 events with exact counts.

ExecutorService: 3+ configurable producers, thread-safe state, graceful shutdown.

Trade-off commits: 066c3d4 adds a Visitor; 08a1dac adds Video and updates all visitors.

GitHub Repository

https://github.com/AlmazCode/SDP-A6